

3.2 Inheritance

Edexcel International GCSE Biology (4BI1) — Paper 2

3.2.1 Key terms

Gene — a length of DNA coding for a protein.

Allele — a different version of a gene.

Genotype — the alleles present for a gene (e.g. Bb).

Phenotype — the observable characteristic (e.g. brown eyes).

Homozygous — two identical alleles (BB or bb).

Heterozygous — two different alleles (Bb).

Dominant (upper-case) — expressed even with one copy.

Recessive (lower-case) — only expressed if homozygous.

Human body cells contain **46** chromosomes (23 pairs) — diploid.

Gametes contain **23** — haploid. Fertilisation restores the diploid number.

3.2.2 Punnett squares

Monohybrid cross Bb × Bb

		Father's gametes	
		B	b
Mother's gametes	B	BB	Bb
	b	Bb	bb

Genotypic ratio:

1 BB : 2 Bb : 1 bb

Phenotypic ratio:

3 dominant : 1 recessive

Probability of recessive trait: 25%.

3.2.3 Sex determination

Males are **XY**, females **XX**. Mother gives X; father gives X or Y. Equal 50:50 chance of either sex.

3.2.4 Mutation

A random change in the DNA base sequence. Can be:

- Neutral (no effect) — most mutations
- Harmful (some)
- Occasionally beneficial — raw material of evolution.

Rate increased by **mutagens**: ionising radiation, UV light, chemicals (tar, mustard gas).

3.2.5 Evolution by natural selection

1. Variation exists in population (mutation + sexual reproduction).
2. Too many offspring — competition.
3. Best-adapted individuals survive to reproduce.
4. They pass on advantageous alleles.
5. Over generations, allele frequency shifts — evolution.

Exam Tip: Antibiotic resistance in bacteria is a classic example. Random mutation gives a resistance allele; antibiotic kills non-resistant; resistant survive and reproduce → allele becomes common. *Never* say bacteria "try" or "want" to become resistant.

3.2.6 Variation

- **Continuous** — many genes + environment, e.g. height, mass. Normal distribution curve.
- **Discontinuous** — single gene, distinct categories, e.g. blood group, tongue rolling. Bar chart.